

Algebra 1
Unit 7
Lesson 3 Practice

Name Answer key
Date _____
Period _____

For questions 1-5, factor each quadratic expression using the method of your choice.

1. $b^2 + 17b + 72$

$(b + 9)(b + 8)$

2. $h^2 - 23h + 132$

$(h - 11)(h - 12)$

3. $x^2 + 15x + 56$

$(x + 7)(x + 8)$

4. $y^2 - 3y - 18$

$(y - 6)(y + 3)$

5. $x^2 - 7x - 30$

$(x - 10)(x + 3)$

For questions 6-7, consider the trinomial $f^2 - 10f + 9$.

6. Complete the area model. $(f - 9)(f - 1)$

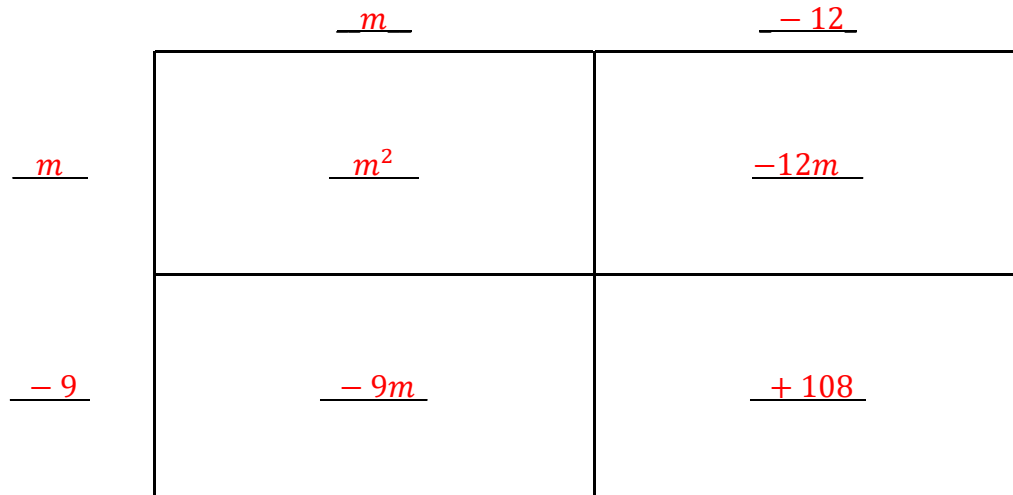
	f	<u>-9</u>
f	f^2	<u>$-9f$</u>
<u>-1</u>	<u>$-f$</u>	9

7. What are the two binomial factors for $f^2 - 10f + 9$?

$$(f - 9)(f - 1)$$

For questions 8-9, consider the trinomial $m^2 - 21m + 108$.

8. Complete the area model.



9. What are the two binomial factors for $m^2 - 21m + 108$?

$$(m - 12)(m - 9)$$

10. Factor $x^2 - 12x + 35$ using factoring by grouping.

$$\begin{aligned} &x^2 - 7x - 5x + 35 \\ &x(x - 7) - 5(x - 7) \\ &(x - 5)(x - 7) \end{aligned}$$